

Yale University

# Latent Variables and EM

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S&DS 265 · Introductory Machine Learning · Lecture 16

Zhuoran Yang

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AI usage: figures were generated, and these slides polished, with the help of AI tools.

# Outline

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1. Latent-Variable Mixture Models
2. The Posterior Assignment
3. Expectation–Maximization Updates
4. The Evidence Lower Bound
5. Checking the Fit, and What Can Go Wrong

# Learning Objectives

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In this lecture you will learn:

1. Why a latent component variable makes a mixture easy to describe but hard to fit;
2. How Bayes' rule turns current parameters into a posterior over the hidden assignment;
3. How weighted complete-data fitting produces the EM parameter updates;
4. Why the evidence lower bound explains EM's monotone likelihood improvement;
5. How restarts, effective counts, and covariance checks reveal mixture failures.

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# Motivating Example: What Kinds of Car Are These?

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You are handed 392 cars, each with a weight and a mileage figure.

What kinds of car are in there, and how common is each kind?

Nothing in the table says which kind a car is.

A **kind** is a group of similar cars; in the model it will be called a component.

Example and data: ISLP Auto data (James et al., 2023).

# The Auto Data

| model                     | mpg | cylinders | horsepower | weight | year |
|---------------------------|-----|-----------|------------|--------|------|
| chevrolet chevelle malibu | 18  | 8         | 130        | 3504   | 70   |
| buick skylark 320         | 15  | 8         | 165        | 3693   | 70   |
| plymouth satellite        | 18  | 8         | 150        | 3436   | 70   |
| amc rebel sst             | 16  | 8         | 150        | 3433   | 70   |

The local Auto table contains 392 complete fuel-economy records. Throughout the lecture the model sees only **weight** and **mpg**; every other column is held back.

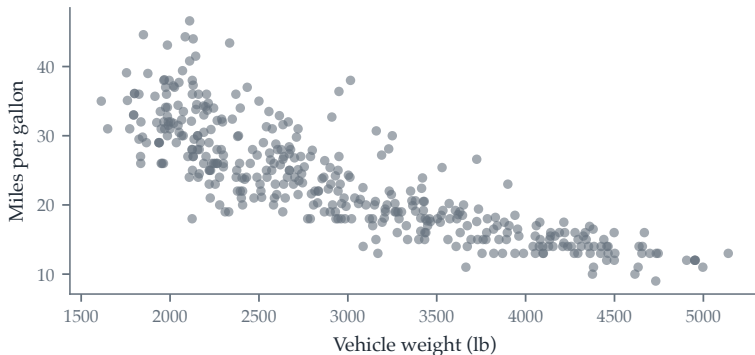
Source: First four complete rows of Auto.csv from the ISLP resources.

## Reading One Row

| model           | mpg | cylinders | horsepower | weight | year |
|-----------------|-----|-----------|------------|--------|------|
| Chevelle Malibu | 18  | 8         | 130        | 3504   | 70   |

This car achieves 18 mpg, has eight cylinders and 130 horsepower, weighs 3,504 pounds, and is from model year 1970. Which class of car it belongs to is a question the row does not answer.

# This Is Everything the Method Is Given

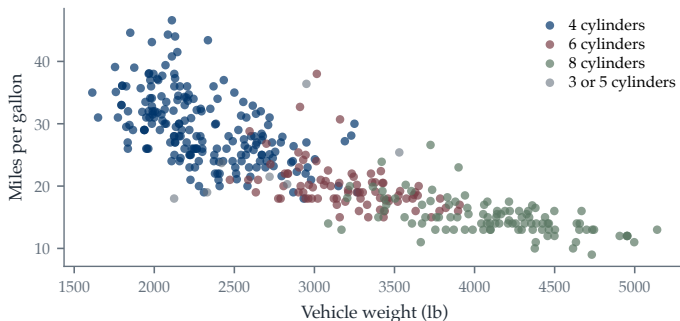


Each point is one car,  $x_i \in \mathbb{R}^2$ . Heavy cars use more fuel, so the cloud slopes down — but nothing in the picture says **group**, and no column in the table does either.

ISLP Auto data, 392 complete cases; weight and fuel economy only.



# This Data Has an Answer Key



Auto happens to record the number of cylinders, and it lines up with the picture: three engine classes in **different regions**, **overlapping where they meet**. We will not let the model see this column — it is how we check the answer afterwards. In most problems no such column exists. ISLP Auto data; cylinders are used to colour this figure only, never to fit a model.

# A Hard Label Overstates What We Know

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Last lecture's answer was one label per car,  $c_i \in \{1, \dots, K\}$ .

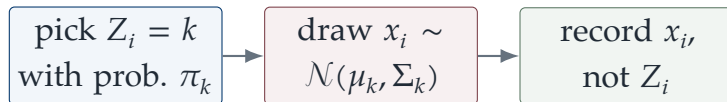
The **Ford Maverick** weighs 3,012 lb and gets 24 mpg, which puts it squarely between two of the groups.  $K$ -means files it under one of them — with the **same confidence** it gives a 1,800 lb compact.

We want the answer to be a **share** instead: this car is about half of one kind and half of the other. A share has to come from somewhere, so each group will need a **spread** and a **prevalence**, not just a center.

# The Model: a Gaussian Mixture

Assume each car comes from one of  $K = 3$  populations. Population  $k$  is Gaussian, and it is chosen with probability  $\pi_k$ :

$$\Pr(Z_i = k) = \pi_k, \quad x_i \mid Z_i = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad \sum_{k=1}^K \pi_k = 1.$$



**$Z_i$  is the assignment:** the kind that produced car  $i$ . The model needs it, the table never contains it, and such a variable is called **latent**.

# What the Three Parameters Mean

- ▶  $\pi_k$  — **how common the kind is**: the share of all cars that are of kind  $k$ . If  $\pi_2 = .26$ , about a quarter of the market is kind 2.
- ▶  $\mu_k$  — **what it looks like**: the typical weight and mileage of kind  $k$ , a single point in the same plane as the data.
- ▶  $\Sigma_k$  — **how much it varies**: how far cars of kind  $k$  spread around  $\mu_k$ , and whether weight and mileage move together inside the kind.

Only  $\mu_k$  has a counterpart in  $K$ -means.  $\pi_k$  and  $\Sigma_k$  are what turn membership into a share.

# The Density That Follows

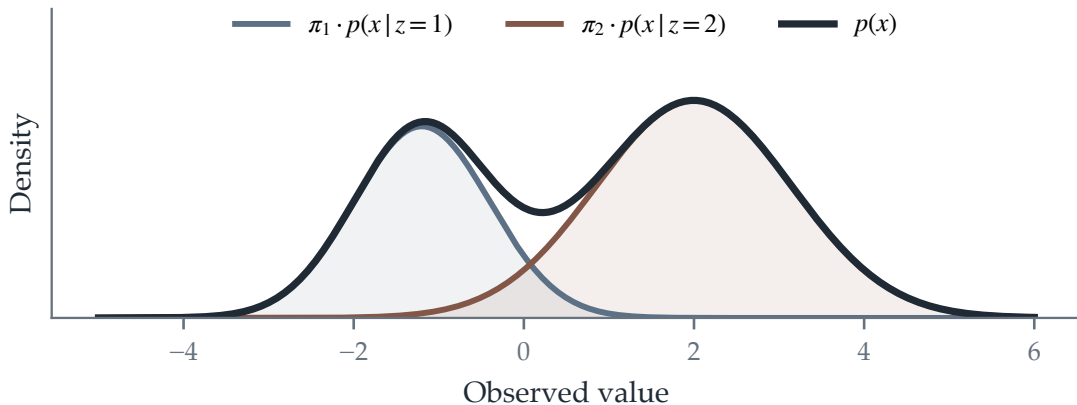
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Averaging over the unseen assignment gives the density of a single car:

$$p(x) = \sum_{k=1}^K \Pr(Z = k) \cdot p(x | Z = k) = \sum_{k=1}^K \pi_k \cdot \mathcal{N}(x; \mu_k, \Sigma_k).$$

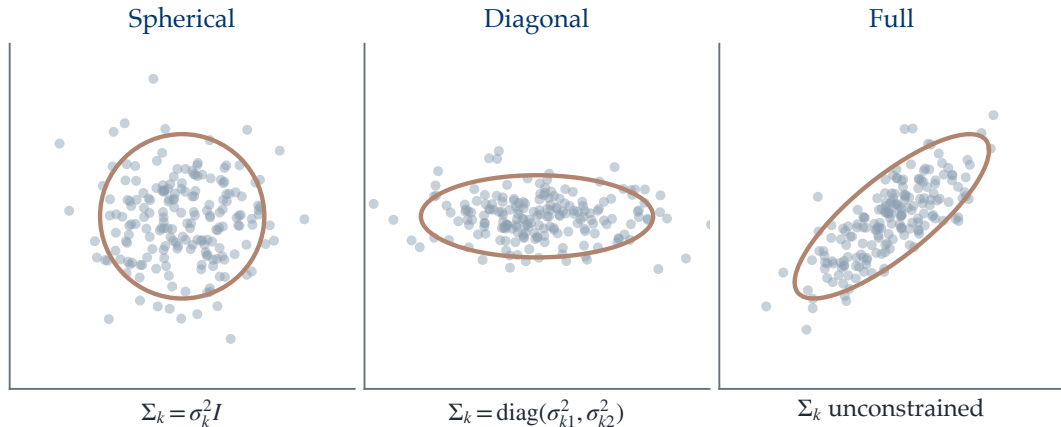
Every kind contributes to every car, weighted by how **common** that kind is.

# A Mixture Adds Component Densities



Weighted component densities sum to the observed density.

# Covariance Assumptions Define Component Geometry



Original Gaussian simulation, seed 26519; red curves are two-standard-deviation ellipses.

# What a Probability Model Buys over a Partition

$K$ -means returns labels and centers. The mixture returns a **density**.

- ▶ **How sure?** Membership is a probability: the Maverick is about half one kind and half another.
- ▶ **How common?**  $\pi_k$  estimates the size of each kind — a summary of the market, not only of the table.
- ▶ **Does the model fit this car?**  $p_\theta(x)$  is **small** exactly when the model finds that car improbable.
- ▶ **Which model fits better?** Adding those up,  $\sum_i \log p_\theta(x_i)$  grades the whole model on held-out cars:  $K = 2$  against  $K = 3$ , spherical against full.



# Where the Three Components Came From

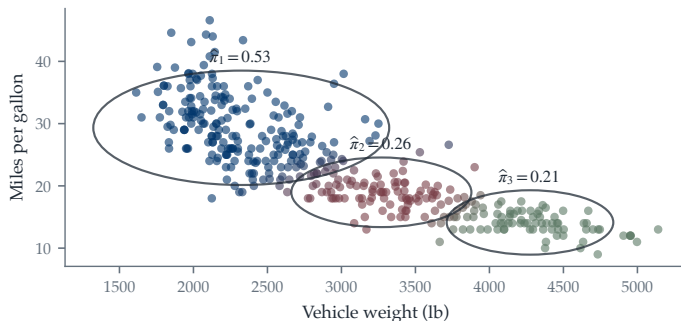
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We did not find three Gaussian populations in the table. We **imposed** them: the number  $K$ , the Gaussian shape, and independence across cars are assumptions, and none of them is checked by the data before we fit.

Two consequences to keep in view:

- ▶ Every conclusion inherits the assumption;
- ▶ Several Gaussians can approximate one non-Gaussian shape, so a fitted component **need not be a real class**. We come back to this at the end, when we finally look at the cylinder column.

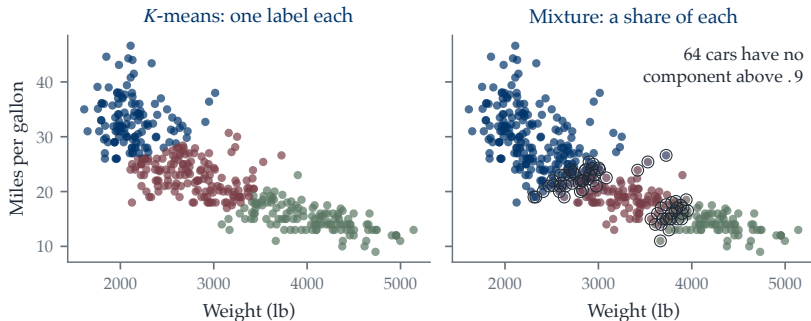
# The Assumption, Fitted to the Cars



Three components: circles are two-standard-deviation contours, colour blends the three memberships, and  $\hat{\pi}_k$  is each group's estimated prevalence. Where did these numbers come from? That question is the rest of the lecture.

ISLP Auto data; course spherical EM, seed 26519, best of twenty starts.

# The Same Cars, Two Kinds of Answer



K-means states a label for every car with identical confidence. The mixture states a share, and 64 of 392 cars have no component above .9 — they sit exactly on the two seams the first picture drew straight through.

ISLP Auto data; K-means and spherical EM on the same standardized features, seed 26519.

# Estimation by Maximum Likelihood

Write  $\theta = (\pi_k, \mu_k, \Sigma_k)_{k=1}^K$  and choose it to make the observed cars as probable as possible:

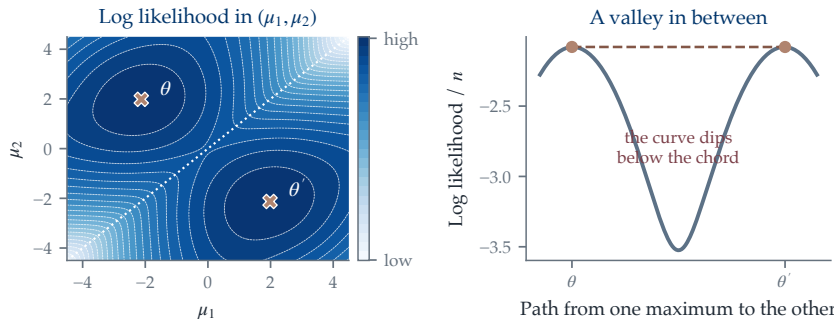
$$\hat{\theta} \in \arg \max_{\theta} \ell(\theta), \quad \ell(\theta) = \sum_{i=1}^n \log p_{\theta}(x_i).$$

Substituting the mixture density,

$$\ell(\theta) = \sum_{i=1}^n \log \left[ \sum_{k=1}^K \pi_k \cdot \mathcal{N}(x_i; \mu_k, \Sigma_k) \right].$$

The **sum sits inside the logarithm**. Nothing separates: every  $\mu_k$  appears in every term, and  $\ell$  is **not concave**.

# Nonconcave Already with Two Means in One Dimension



Two components, unit variances, equal weights: only  $\mu_1$  and  $\mu_2$  are free. Relabelling maps  $\theta$  to  $\theta'$  without changing the likelihood, so there are **two separated maxima** — and the straight path between them runs through a valley. A concave function cannot do that.

Original simulation, seed 26519; 240 draws from an equal mixture of  $\mathcal{N}(-2, 1)$  and  $\mathcal{N}(2, 1)$ .

# The General Latent-Variable Model

A latent-variable model specifies a joint distribution over the hidden  $z$  and the observed  $x$ ,

$$p_{\theta}(x, z) = p_{\theta}(z) p_{\theta}(x | z),$$

with both factors chosen to be simple. The mixture is the case  $p_{\theta}(z = k) = \pi_k$  and  $p_{\theta}(x | z = k) = \mathcal{N}(x; \mu_k, \Sigma_k)$ .

Only  $x$  is recorded, so maximum likelihood must fit the **marginal**:

$$\ell(\theta) = \sum_{i=1}^n \log p_{\theta}(x_i) = \sum_{i=1}^n \log \sum_z p_{\theta}(x_i, z).$$

The logarithm never reaches the simple factors, and  $\ell$  is **nonconcave** for the same reason as before.

# Idea: Keep the Alternation, Soften the Assignment

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## *K*-means alternated

1. Send each point to its nearest center;
2. Recompute each center from the points sent to it.

## We alternate

1. **Guess the assignment**: holding  $\theta$  fixed, Bayes' rule turns each car into a **share**  $\gamma_{ik}$ ;
2. Refit  $\pi, \mu, \Sigma$ , counting every car as **partly** in each kind.

Step 2 is tractable: with the shares held fixed, the logarithm sits on **one Gaussian at a time** again, and each component has a closed-form weighted fit.

# What Remains

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- ▶ **Next — the assignment.** Bayes' rule turns the current parameters into  $\gamma_{ik} = \Pr(Z_i = k \mid x_i)$ .
- ▶ **Then — the parameters.** Weighted means, weighted covariances, and  $\pi_k$  as a fractional count.
- ▶ **Then — why it works.** The two steps are climbing a **lower bound** on  $\ell(\theta)$ , the evidence lower bound, which is why the likelihood never decreases.

Together the two steps are the expectation–maximization algorithm.



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# The Model Already Supplies a Prior and a Likelihood

Fix the parameters  $\theta$  at their current values. The mixture then states two things about a single car.

**Prior** — before seeing  $x_i$

$$\Pr(Z_i = k) = \pi_k$$

how common kind  $k$  is.

**Likelihood** — if kind  $k$  produced it

$$p(x_i | Z_i = k) = \mathcal{N}(x_i; \mu_k, \Sigma_k)$$

how plausible  $x_i$  looks under kind  $k$ .

What we want is the reverse question: **given** that we observed  $x_i$ , which kind produced it?

# Computing the Posterior

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Bayes' rule combines them — posterior  $\propto$  prior  $\times$  likelihood, normalized over the  $K$  kinds:

$$\gamma_{ik} = \Pr(Z_i = k \mid x_i) = \frac{\Pr(Z_i = k) \cdot p(x_i \mid Z_i = k)}{\sum_{\ell=1}^K \Pr(Z_i = \ell) \cdot p(x_i \mid Z_i = \ell)},$$

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which for the Gaussian mixture is

$$\gamma_{ik} = \frac{\pi_k \cdot \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_{\ell=1}^K \pi_{\ell} \cdot \mathcal{N}(x_i; \mu_{\ell}, \Sigma_{\ell})}.$$

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The denominator is  $p_{\theta}(x_i)$ , the mixture density from the last section.

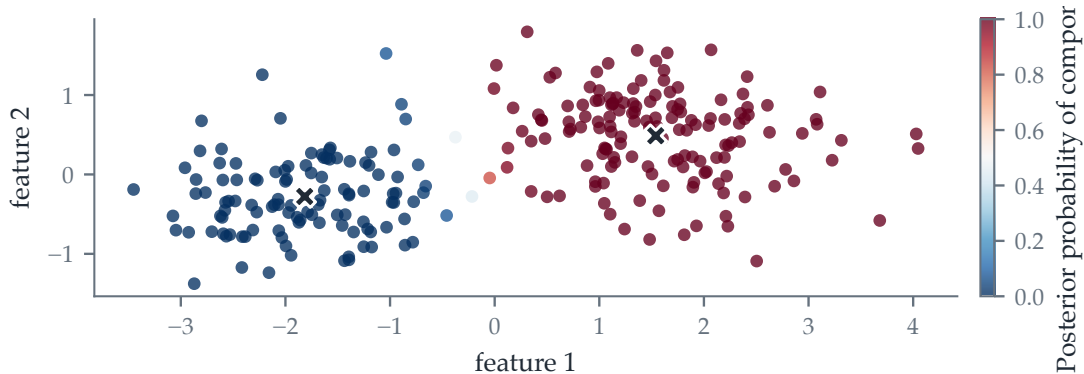
# What $\gamma_{ik}$ Means

$\gamma_{ik}$  is a **posterior probability**: given the evidence  $x_i$ , how likely it is that car  $i$  came from kind  $k$ .

- ▶ Before seeing the car, kind  $k$  had probability  $\pi_k$ . After seeing its weight and mileage, that becomes  $\gamma_{ik}$ : the evidence has **reweighted** the kinds, and  $\sum_k \gamma_{ik} = 1$  still holds.
- ▶ It is **conditional on  $\theta$** . Change the parameters and every posterior changes —  $\gamma_{ik}$  is the model's current belief, not a fact about the car.
- ▶ A hard label is the extreme case  $\gamma_{ik} \in \{0, 1\}$ , and  $K$ -means reports nothing else.

Many textbooks call  $\gamma_{ik}$  the **responsibility** of component  $k$  for observation  $i$ .

# The Posterior Preserves Ambiguity



Original course simulation, seed 26519; colour is the posterior probability of component 2.

# One Posterior Calculation

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Suppose equal priors, unit variances, means 0 and 3, and observation  $x = 1$ . The common Gaussian constant cancels:

$$w_1 = \frac{1}{2}e^{-(1-0)^2/2} = \frac{1}{2}e^{-1/2}, \quad w_2 = \frac{1}{2}e^{-(1-3)^2/2} = \frac{1}{2}e^{-2}.$$

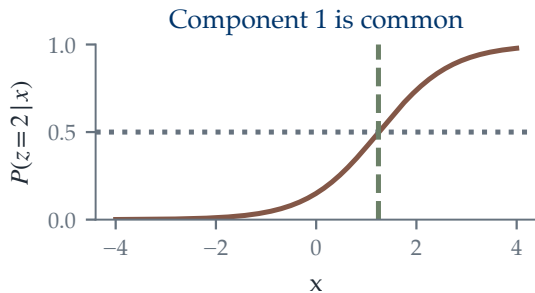
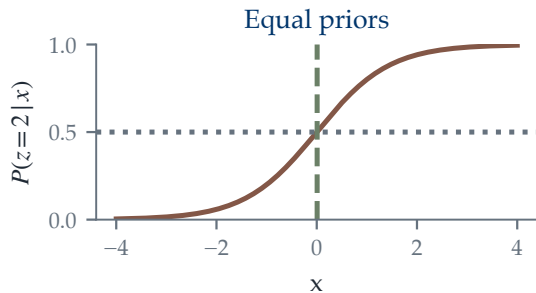
Therefore

$$\gamma_1 = \frac{w_1}{w_1 + w_2} = \frac{e^{-1/2}}{e^{-1/2} + e^{-2}} \approx .818, \quad \gamma_2 \approx .182.$$

A hard nearest-mean rule would discard this uncertainty.



# Component Priors Shift Posterior Membership



Only the component prior weights differ.

# Mean Distance Is Only One Factor

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Taking logarithms of the posterior numerator gives

$$\log \pi_k - \frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k) + \text{constant}.$$

- ▶  $\pi_k$  favors common components;
- ▶  $|\Sigma_k|$  controls density concentration;
- ▶ Mahalanobis distance uses component-specific shape.

Thus the posterior is not decided by nearest Euclidean mean alone.

# Stable Log-Sum-Exp Calculation



Define  $a_{ik} = \log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k)$  and  $m_i = \max_k a_{ik}$ . Then

$$\begin{aligned}\log \sum_k e^{a_{ik}} &= \log \left[ e^{m_i} \sum_k e^{a_{ik} - m_i} \right] \\ &= m_i + \log \sum_k e^{a_{ik} - m_i}.\end{aligned}$$

Every exponent  $a_{ik} - m_i \leq 0$ , preventing large exponentials and reducing underflow before normalization.

## Next: the Other Half of the Loop

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With  $\theta$  in hand we can compute every  $\gamma_{ik}$ . The remaining question is the reverse one:

given the posteriors, how do we re-estimate  $\pi_k$ ,  $\mu_k$ , and  $\Sigma_k$ ?

Next we write the likelihood we would have had if the assignments were observed, and shows that replacing each label by its  $\gamma_{ik}$  makes that fit **a weighted version of the ordinary Gaussian estimates**.

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# Complete-Data Log Likelihood

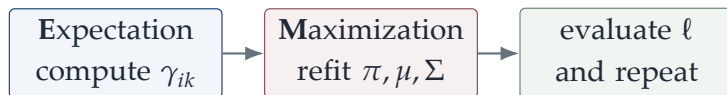
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If labels were observed, write  $z_{ik} = \mathbf{1}\{Z_i = k\}$ . Then

$$\ell_c(\theta) = \sum_{i=1}^n \sum_{k=1}^K z_{ik} [\log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k)].$$

Conditional on  $z$ , parameters separate by component. EM replaces the missing indicator by its posterior expectation and repeats.

# One Cycle: Expectation, then Maximization



The two letters name what each half does. The **E-step** takes the **expectation** of  $\ell_c(\theta)$  over the unknown assignments, using the posterior at the current parameters. The **M-step** **maximizes** that expected likelihood over  $\theta$ .

# The E-Step: an Expectation over the Unknown $Z$

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We cannot evaluate  $\ell_c(\theta)$ , because  $z_{ik}$  is not observed. Replace each indicator by its expected value under the current posterior:

$$\mathbb{E}[z_{ik} \mid x_i, \theta^{(t)}] = \Pr(Z_i = k \mid x_i, \theta^{(t)}) = \gamma_{ik}^{(t)}.$$

Carrying that through  $\ell_c$  gives the objective the M-step will maximize:

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{Z \mid X, \theta^{(t)}}[\ell_c(\theta)] = \sum_{i,k} \gamma_{ik}^{(t)} [\log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k)].$$

Every fractional assignment is kept; no hard label is imputed.



# The M-Step: Maximize $Q$ over $\theta$



Define the effective count  $N_k = \sum_i \gamma_{ik}$ . Maximizing  $\sum_k N_k \log \pi_k$  subject to  $\sum_k \pi_k = 1$  gives

$$\pi_k^{\text{new}} = \frac{N_k}{n}.$$

For fixed covariance, differentiating the weighted quadratic yields

$$0 = \sum_i \gamma_{ik}(x_i - \mu_k) \quad \Rightarrow \quad \boxed{\mu_k^{\text{new}} = \frac{1}{N_k} \sum_i \gamma_{ik} x_i}.$$

$N_k$  plays the role of a fractional sample size.

# Covariance Update

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The weighted covariance is

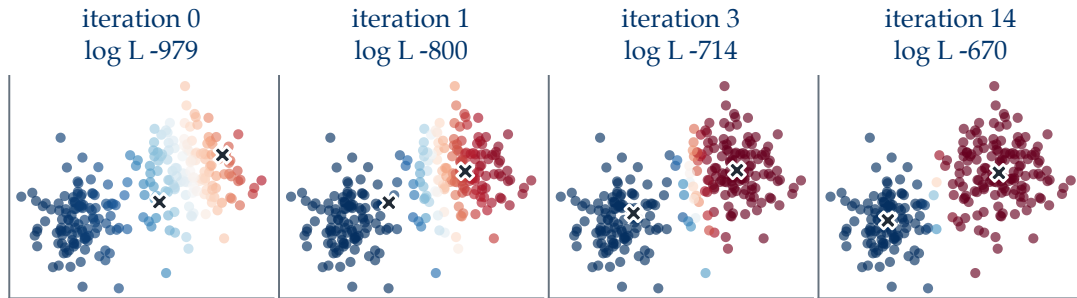
$$\Sigma_k^{\text{new}} = \frac{1}{N_k} \sum_{i=1}^n \gamma_{ik} (x_i - \mu_k^{\text{new}})(x_i - \mu_k^{\text{new}})^\top.$$

A diagonal family keeps only diagonal entries. For spherical covariance in  $d$  dimensions,

$$(\sigma_k^2)^{\text{new}} = \frac{1}{dN_k} \sum_i \gamma_{ik} \|x_i - \mu_k^{\text{new}}\|_2^2.$$

Covariance constraints trade geometric flexibility for statistical stability.

# Posteriors and Means Co-Evolve



Original spherical-mixture simulation; colour is the posterior for the second component and crosses are means.

# One M-Step Calculation

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For  $x = (0, 1, 4)$  and component-1 posteriors  $(.9, .8, .1)$ ,

$$N_1 = .9 + .8 + .1 = 1.8, \quad \pi_1^{\text{new}} = 1.8/3 = .6,$$

$$\mu_1^{\text{new}} = \frac{.9(0) + .8(1) + .1(4)}{1.8} = \frac{1.2}{1.8} = .667.$$

Complementary posteriors give  $N_2 = 1.2$  and  $\pi_2^{\text{new}} = .4$ . The component means are weighted, not based on thresholded labels.

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# A Lower Bound on the Log Likelihood

Take **any** distribution  $q(z)$  over the hidden variable, and multiply and divide by it inside the same sum as before:

$$\begin{aligned}\log p_{\theta}(x) &= \log \sum_z p_{\theta}(x, z) = \log \sum_z q(z) \cdot \frac{p_{\theta}(x, z)}{q(z)} \\ &\geq \sum_z q(z) \cdot \log \frac{p_{\theta}(x, z)}{q(z)} \equiv \mathcal{L}(q, \theta).\end{aligned}$$

$$\boxed{\log p_{\theta}(x) \geq \mathcal{L}(q, \theta) \quad \text{for every } q}$$

$\mathcal{L}$  is the **evidence lower bound**, or ELBO. The logarithm has moved **inside** the sum, so it now sits on the joint  $p_{\theta}(x, z)$ , which factorizes.

Source: The inequality is Jensen's; derived in Appendix A.1.

# Inside the Bound, the Joint Splits

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The joint is prior times likelihood,  $p_{\theta}(x, z) = p_{\theta}(z) \cdot p_{\theta}(x | z)$ , and the logarithm now reaches both factors:

$$\mathcal{L}(q, \theta) = \sum_z q(z) [\log p_{\theta}(z) + \log p_{\theta}(x | z)] - \sum_z q(z) \log q(z).$$

# Inside the Bound, the Joint Splits

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The second sum has **no  $\theta$  in it**, so it is a constant as far as the M-step is concerned:

$$\arg \max_{\theta} \mathcal{L}(q, \theta) = \arg \max_{\theta} \sum_z q(z) \log p_{\theta}(x, z) = \arg \max_{\theta} \mathbb{E}_q[\log p_{\theta}(x, z)].$$

For the mixture this is the  $Q$  already derived, with  $q(k)$  in place of  $\gamma_{ik}$ : maximizing the bound over  $\theta$  is the weighted fit already derived.



# The Same Objects, in the Gaussian Mixture

|                   | general latent-variable model             | Gaussian mixture, car $i$                              |
|-------------------|-------------------------------------------|--------------------------------------------------------|
| prior             | $p_{\theta}(z)$                           | $\Pr(Z_i = k) = \pi_k$                                 |
| likelihood        | $p_{\theta}(x   z)$                       | $\mathcal{N}(x_i; \mu_k, \Sigma_k)$                    |
| joint             | $p_{\theta}(x, z)$                        | $\pi_k \cdot \mathcal{N}(x_i; \mu_k, \Sigma_k)$        |
| marginal          | $p_{\theta}(x) = \sum_z p_{\theta}(x, z)$ | $\sum_k \pi_k \cdot \mathcal{N}(x_i; \mu_k, \Sigma_k)$ |
| posterior         | $p_{\theta}(z   x)$                       | $\gamma_{ik}$                                          |
| free distribution | $q(z)$ , any distribution                 | $(q_{i1}, \dots, q_{iK})$ with $\sum_k q_{ik} = 1$     |

The bound holds for **every** choice of  $q_i$ . The next slide picks the one that makes it tight.

# EM in the Abstract Form: Alternating Optimization on ELBO

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$\mathcal{L}(q, \theta)$  has two arguments, so maximize it in one and then the other:

$$q^{(t+1)} = \arg \max_q \mathcal{L}(q, \theta^{(t)}), \quad \theta^{(t+1)} = \arg \max_{\theta} \mathcal{L}(q^{(t+1)}, \theta).$$

The second is the weighted maximum-likelihood fit already derived. The first looks harder — it optimizes over **distributions**, not numbers.

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The second is the weighted maximum-likelihood fit already derived. The first looks harder — it optimizes over **distributions**, not numbers.

**It has a closed-form answer.**

# The Best $q$ Is the Posterior

Write  $D_{\text{KL}}(q \| p) = \sum_z q(z) \log \frac{q(z)}{p(z)}$  for the **Kullback–Leibler divergence**: it is **never negative** and is zero exactly when  $q = p$ . Then, for every  $q$ ,

$$\log p_{\theta}(x) = \mathcal{L}(q, \theta) + D_{\text{KL}}(q \| p_{\theta}(z | x))$$

The left side does not involve  $q$ , so maximizing  $\mathcal{L}$  over  $q$  is the same as driving that gap to zero:

$$q^{(t+1)}(z) = p_{\theta^{(t)}}(z | x), \quad \mathcal{L}(q^{(t+1)}, \theta^{(t)}) = \log p_{\theta^{(t)}}(x).$$

That is the E-step, and for the mixture this posterior is  $(\gamma_{i1}, \dots, \gamma_{iK})$ .

Source: Identity derived in Appendix A.2.

# The M-Step Raises the Bound

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Hold  $q^{(t+1)}$  fixed and choose  $\theta^{(t+1)}$  to increase the ELBO:

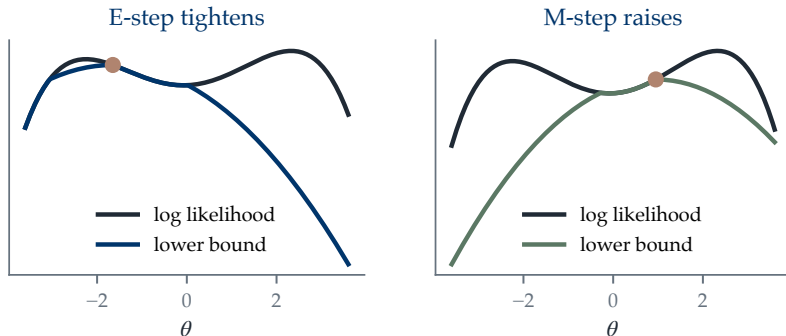
$$\mathcal{L}(q^{(t+1)}, \theta^{(t+1)}) \geq \mathcal{L}(q^{(t+1)}, \theta^{(t)}).$$

Since the log likelihood is always at least its ELBO,

$$\log p_{\theta^{(t+1)}}(x) \geq \mathcal{L}(q^{(t+1)}, \theta^{(t+1)}).$$

Combining these inequalities proves observed-likelihood monotonicity.

# Tighten the Bound, Then Raise It



Source: Original pedagogical schematic; not a fitted-mixture trace.

# Why EM Cannot Go Downhill

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Chaining the two steps gives

$$\log p_{\theta^{(t+1)}}(x) \geq \log p_{\theta^{(t)}}(x)$$

for every iteration: the observed likelihood **never decreases**.

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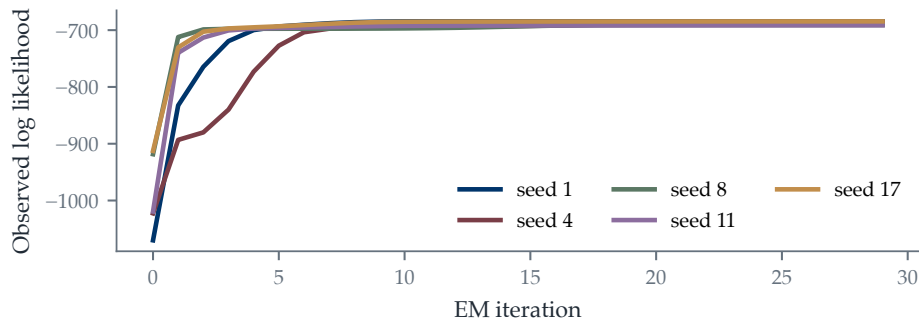
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for every iteration: the observed likelihood **never decreases**. The E-step makes the bound touch the likelihood, so raising the bound in the M-step must carry the likelihood with it.

**Monotone is not optimal. It guarantees no global maximum, no unique parameters, and no meaningful interpretation of a component — only that each step is not a step backwards.**

Source: Three-line chain in Appendix A.4.

# Monotone Restarts Can End Differently



Every run climbs, and they do not all arrive at the same place. Compare runs by their [final likelihood](#), never by component number: relabelling the components leaves the likelihood unchanged, so “component 2” in one run has nothing to do with “component 2” in another. Original three-component simulation with five initializations.

# Outline

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1. Latent-Variable Mixture Models
2. The Posterior Assignment
3. Expectation–Maximization Updates
4. The Evidence Lower Bound
5. Checking the Fit, and What Can Go Wrong

# Back to the Answer Key

The three fitted components, in the original units, beside the withheld cylinder column:

| component | $\hat{\pi}_k$ | weight | mpg | 4 cyl | 6 cyl | 8 cyl |
|-----------|---------------|--------|-----|-------|-------|-------|
| 1         | .53           | 2,323  | 29  | 190   | 13    | 1     |
| 2         | .26           | 3,270  | 19  | 9     | 65    | 22    |
| 3         | .21           | 4,274  | 14  | 0     | 5     | 80    |

The model never saw this column, yet the largest-posterior assignment reproduces it for **335 of 385** cars; the 50 disagreements sit in the overlap.

Source: ISLP Auto data; seven 3- or 5-cylinder cars omitted.

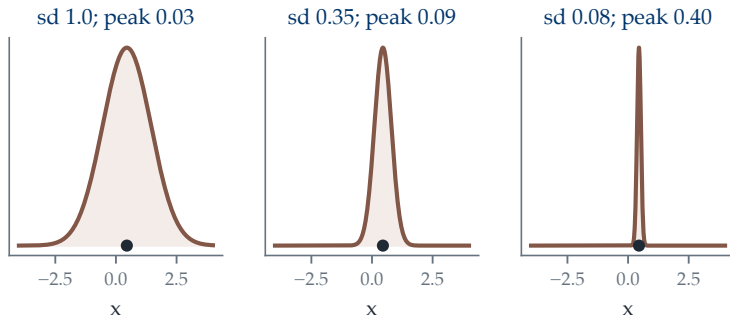
# Here We Could Check. Usually We Cannot

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On this data the components track a real engineering distinction rather than an arbitrary cut through a smooth cloud — and we know that only because a **column existed** to compare against.

**Most problems have no such column.** The remaining slides are the diagnostics that work without one: what goes wrong, and how it shows itself in the fit rather than in an answer key.

# A Component Can Collapse onto One Observation



This is the degenerate case where a component ends up with effectively **one data point** —  $N_k \rightarrow 1$  — so its mean sits exactly on that point, the weighted spread around it is zero, and its density there grows without bound.

Original one-dimensional calculation; component weight is .08.

# Why Gaussian-Mixture Likelihood Can Diverge



For one component centered exactly at observation  $x_i$ , a  $d$ -dimensional spherical density is

$$\mathcal{N}(x_i; x_i, \sigma^2 I) = (2\pi\sigma^2)^{-d/2}.$$

As  $\sigma^2 \rightarrow 0$ ,

$$(2\pi\sigma^2)^{-d/2} \rightarrow \infty, \quad \log p_\theta(x_i) \rightarrow \infty.$$

Collapse is a property of the unregularized likelihood, not only floating-point arithmetic.

# Preventing Degenerate Components

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## Change the fitted problem

variance floors, tied covariance,  
penalties, or priors prevent arbitrarily  
sharp components

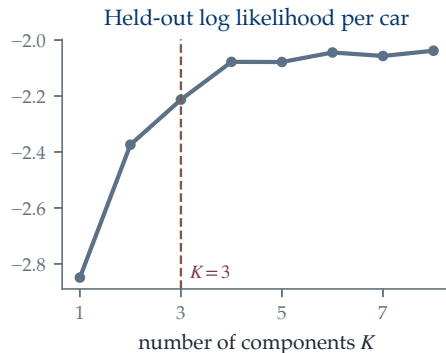
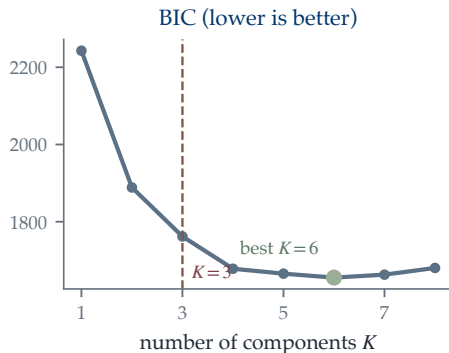
## Inspect the result

multiple starts, effective counts  $N_k$ ,  
held-out likelihood, and density plots  
reveal fragile fits

Regularization should be declared as part of the model, not hidden as a numerical patch.



# Choosing $K$ on the Cars



Both curves improve steeply to  $K = 3$  or  $4$  and then flatten — and BIC's actual minimum is at  $K = 6$ . Past the elbow the extra components are **tiling a curved band**, not finding new kinds of car.

ISLP Auto data; course spherical EM, best of ten starts per  $K$ ; five-fold split, seed 26519.

# A Rule of Thumb for $K$

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- ▶ Fit  $K = 1, 2, \dots$  and plot both curves. Take the **elbow**, the smallest  $K$  after which the gain is small — not the strict optimum.
- ▶ Check the smallest effective count  $N_k = \sum_i \gamma_{ik}$ : here it falls from 82 at  $K = 3$  to 14 at  $K = 8$ . A component holding a handful of points is an artifact, and a collapse waiting to happen.
- ▶ **BIC** =  $-2\ell + p \log n$  charges for parameters ( $p = 4K - 1$  here; see Appendix A.3); **held-out likelihood** charges nothing but needs a split.

A good density and a meaningful set of kinds are different goals, and they do not always want the same  $K$ .

# From Exact EM to Variational Inference

---

## Exact E-step

compute  $p_{\theta}(z | x)$  in closed form and  
make the ELBO tight

## Approximate inference

choose  $q_{\phi}(z | x)$  from a tractable family  
and optimize an approximate ELBO

# From Exact EM to Variational Inference

## Exact E-step

compute  $p_{\theta}(z | x)$  in closed form and  
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## Approximate inference

choose  $q_{\phi}(z | x)$  from a tractable family  
and optimize an approximate ELBO

Two things break when the model is a neural network:  $p_{\theta}(z | x)$  has no closed form, and recomputing it per observation is too slow. A **variational autoencoder** answers both by **learning** the E-step — one encoder network  $q_{\phi}(z | x)$ , shared across all observations, trained jointly with  $\theta$  by maximizing the same ELBO.

**The objective in Lecture 17 is the bound on this slide. Only the way the E-step is obtained changes.**

# Summary: Gaussian Mixtures and EM

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$$\gamma_{ik} = \frac{\pi_k p_k(x_i)}{\sum_{\ell} \pi_{\ell} p_{\ell}(x_i)}, \quad \mu_k^{\text{new}} = \frac{\sum_i \gamma_{ik} x_i}{\sum_i \gamma_{ik}}.$$

1. Mixtures model a density through latent component labels and soft membership.
2. EM alternates posterior inference with weighted maximum likelihood.
3. Priors and covariance geometry matter in addition to mean distance.

## Summary: Bounds and Failure Modes

---

$$\log p_{\theta}(x) = \mathcal{L}(q, \theta) + D_{\text{KL}}(q \| p_{\theta}(z | x)).$$

1. The E-step tightens the ELBO and the M-step raises it, so likelihood does not decrease.
2. Local maxima and covariance collapse require explicit safeguards; component numbers are arbitrary and carry none.
3. A component is model-conditional and need not be a real or causal class.

# Next Lecture

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EM recomputes the posterior separately for every observation, and it needs that posterior in closed form. Neither survives once the model that generates  $x$  from  $z$  is a neural network.

**Lecture 17 — Autoencoders and Variational Autoencoders** replaces the E-step by a single encoder network that outputs an approximate posterior for any input, and trains it against the bound derived today.

Reading: DL Chapter 14; Kingma and Welling (2019), *An Introduction to Variational Autoencoders*.

## A.1 The Jensen Lower Bound



Insert any distribution  $q(z)$  whose support covers the joint model, then apply Jensen's inequality to the concave logarithm:

$$\begin{aligned}\log p_{\theta}(x) &= \log \sum_z p_{\theta}(x, z) = \log \sum_z q(z) \frac{p_{\theta}(x, z)}{q(z)} \\ &\geq \sum_z q(z) \log \frac{p_{\theta}(x, z)}{q(z)} \equiv \mathcal{L}(q, \theta).\end{aligned}$$

Equality holds when  $p_{\theta}(x, z)/q(z)$  is constant in  $z$ , which forces  $q(z) = p_{\theta}(z | x)$ .



## A.2 The Bound Plus the KL Gap



Splitting the joint as  $p_{\theta}(x, z) = p_{\theta}(z | x) p_{\theta}(x)$  inside the bound gives

$$\begin{aligned}\mathcal{L}(q, \theta) &= \sum_z q(z) \log \frac{p_{\theta}(z | x) p_{\theta}(x)}{q(z)} \\ &= \log p_{\theta}(x) - \sum_z q(z) \log \frac{q(z)}{p_{\theta}(z | x)} = \log p_{\theta}(x) - D_{\text{KL}}(q \| p_{\theta}(z | x)).\end{aligned}$$

Because  $D_{\text{KL}} \geq 0$  with equality only at  $q = p_{\theta}(z | x)$ , the bound is tight exactly at the posterior, which is the E-step.

## A.3 The Bayesian Information Criterion

Rather than hold data out, **BIC** charges the training fit for the parameters it used:

$$\text{BIC} = -2 \ell(\hat{\theta}) + p \log n, \quad \text{smaller is better,}$$

where  $p$  counts free parameters and  $n$  counts observations (here, 392 cars).

- ▶ A spherical mixture in  $d$  dimensions has  $p = K(d + 1) + (K - 1)$ : a mean and a variance per component, plus  $K - 1$  weights. At  $d = 2$ ,  $p = 4K - 1$ .
- ▶ Each parameter costs  $\log n$ , so more data means an extra component must earn more before it is worth keeping.
- ▶ It approximates a Bayesian model comparison under conditions mixtures do not strictly satisfy: a guide, not a test.

Source: Read ISLP Section 6.1.3, “Choosing the Optimal Model” (pp. 235–239).

## A.4 The Monotonicity Chain

The complete argument is

$$\begin{aligned}\log p_{\theta^{(t+1)}}(x) &\geq \mathcal{L}(q^{(t+1)}, \theta^{(t+1)}) \\ &\geq \mathcal{L}(q^{(t+1)}, \theta^{(t)}) \\ &= \log p_{\theta^{(t)}}(x).\end{aligned}$$

This guarantees nondecreasing observed likelihood. It does not guarantee a global maximum, unique parameters, or a scientifically meaningful component interpretation.